EVA: Efficient Reinforcement Learning for End-to-End Video Agent

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SenseTime Research

Abstract

Video understanding with multimodal large language models (MLLMs) remains challenging due to the long token sequences of videos, which contain extensive temporal dependencies and redundant frames. Existing approaches typically treat MLLMs as passive recognizers, processing entire videos or uniformly sampled frames without adaptive reasoning. Recent agent-based methods introduce external tools, yet still depend on manually designed workflows and perception-first strategies, resulting in inefficiency on long videos. We present EVA, an Efficient Reinforcement Learning framework for End-to-End Video Agent, which enables planning-before-perception through iterative summary–plan–action–reflection reasoning. EVA autonomously decides what to watch, when to watch, and how to watch, achieving query-driven and efficient video understanding. To train such agents, we design a simple yet effective three-stage learning pipeline—comprising supervised fine-tuning (SFT), Kahneman–Tversky Optimization (KTO), and Group Relative Policy Optimization (GRPO)—that bridges supervised imitation and reinforcement learning. We further construct high-quality datasets for each stage, supporting stable and reproducible training. We evaluate EVA on six video understanding benchmarks, demonstrating its comprehensive capabilities. Compared with existing baselines, EVA achieves a substantial improvement of 6–12% over general MLLM baselines and a further 1–3% gain over prior adaptive agent methods.

1. Introduction

Video understanding has emerged as a cornerstone of multimodal intelligence [1, 46], enabling a wide range of applications such as video question answering, retrieval, and embodied perception. As multimodal large language models (MLLMs) become increasingly capable of integrating vision, language, and reasoning [13], a new research frontier is opening up—transforming them from passive perception models into active agents. This naturally raises a fundamental question: *How can an MLLM-based Agent decide when and how to watch a video autonomously?*

Most existing video understanding systems still treat MLLMs as passive recognizers—they process entire videos or uniformly sampled frames to generate responses, without any notion of selective attention or adaptive reasoning [18, 32, 46], as illustrated in Figure 1. Recent agent-based approaches take a step forward by introducing external tools such as frame-selection modules [17, 27, 36]. However, these pipelines remain largely handcrafted—built upon fixed parameters, rigid workflows, and limited exploration capabilities (e.g., fixed sampling rates). Moreover, even these "agent-based" methods typically start their reasoning after being fed a set of uniformly sampled frames together with the textual query, making them still perception-first rather than truly planning-driven, which results in redundant visual processing and limited reasoning efficiency on long videos.

To bridge this gap, we advocate a **planning-before-perception** paradigm, where the agent first reasons solely from the textual query to decide what to watch, when to watch, and how to watch, before engaging with any visual input. We formulate such video understanding as an iterative process of *summary–planning–action–reflection*. This paradigm allows the agent to progressively refine its perception and reasoning in response to the query, selectively attending to informative moments while avoiding unnecessary computation. Through this lens, an MLLM evolves from a passive video recognizer into an active, adaptive, and autonomous agentic watcher.

At the heart of our approach lies an **iterative perception, reasoning, and tool usage paradigm** that couples visual summary, planning, tool calling, and reflection thinking. The central challenge is to enable the model to operate effectively within this reasoning loop: learning how to generate the initial tool call based solely on the query without watching the video, how to continue reasoning when

*Equal contribution ‡Project Lead †Corresponding author: luotto@sensetime.com Our Code and model are at this link.

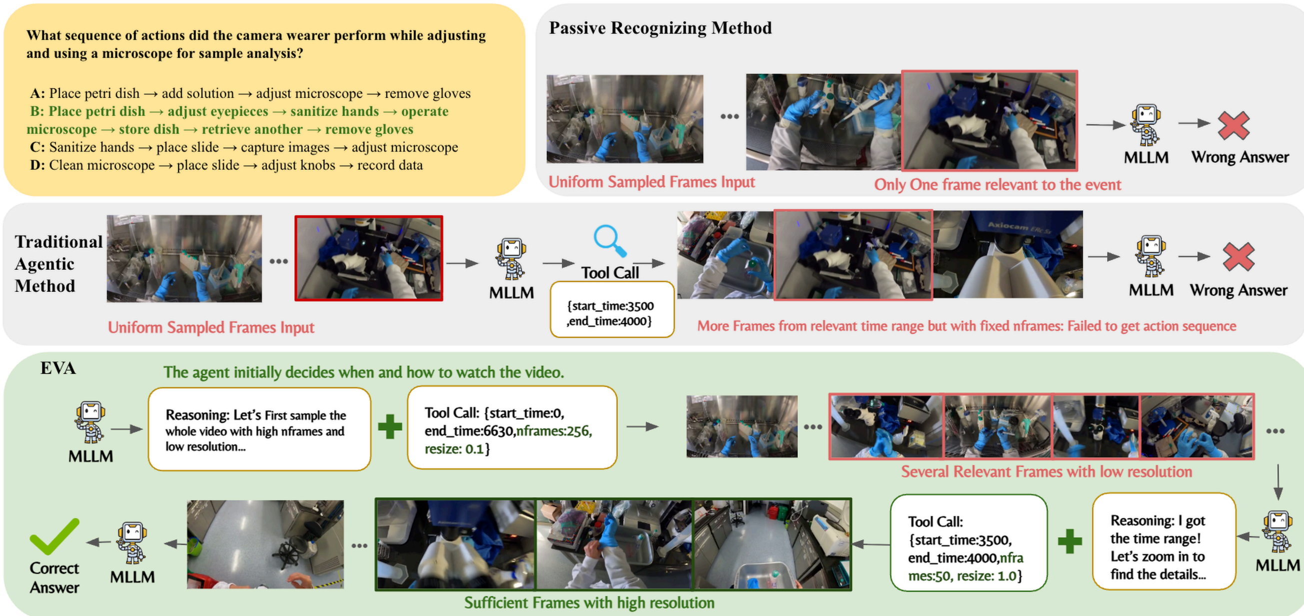

Figure 1. Given a question that requires the MLLM to figure out action sequences from an extremely long video (over 6600 seconds), the traditional uniformed sampling method is limited by the content length of the MLLM, and it is extremely hard for it to sample all the keyframes to answer the question correctly. As for the Traditional Agentic Method, the agent will also be given the uniformly sampled frames along with the video, which already occupy a lot of context. Although the agent can call tools to extract frames from a specific time range, the tool is rigid and the agent cannot adjust the fps and resolution, which leads to potential information loss. However, in EVA, the agent can arrange the tokens wisely. It can first watch the whole video with low resolution and high fps to get an overview of the video without costing too many visual tokens. After it finds the key time range, it will extract frames with high fps and high resolution, which leads to the correct answer.

the available visual information is insufficient, and how to avoid over-exploration or becoming trapped in unnecessary iterations.

To address this challenge, we introduce a three-stage training strategy. In the initial training stage, we construct a **Supervised Fine-Tuning (SFT) Cold-Start** dataset to instill core video-agent capabilities: tool-call formatting, interleaved image–text reasoning, frame-level understanding, and basic frame-selecting strategies. This cold-start phase supplies the model with a stable behavioral prior that eases later, more aggressive optimization. The second stage uses a **Kahneman–Tversky Optimization (KTO)** [10] dataset composed of both successful and failed strategy trajectories. KTO guides the agent to prefer effective strategies while avoiding common failure modes; by correcting these known bad cases prior to GRPO [31], it improves convergence, robustness, and stability during online policy optimization. Third, we introduce an online reinforcement learning stage based on **Group Relative Policy Optimization (GRPO)**, which employs multiple data-driven rewards for both open-ended and multiple-choice question answering (QA). These standard yet flexible reward signals balance reasoning depth with computational efficiency, enabling scalable and adaptive policy learning for video understanding.

Together, these mechanisms enable the agent to learn an adaptive policy for multi-round perception, planning, and tool usage, ensuring effective video understanding while controlling redundant computation.

To ensure stable and reproducible agentic reinforcement learning, we curate and construct a series of high-quality datasets for both the SFT cold-start and reinforcement learning stages. Specifically, we introduce the EVA-SFT, EVA-KTO, and EVA-RL datasets. The EVA-SFT comprises 10k high-quality samples covering both general and task-specific agent training data. The EVA-KTO includes 11k labeled frame-selection strategies, capturing diverse success and failure trajectories to guide strategy optimization. The EVA-RL contains 9.6k open-ended video QA pairs and 1.1k multiple choice questions.

Overall, our main contributions are summarized as follows:

- **A novel and efficient RL-based video agent (EVA).** We propose a planning-before-perception framework featuring iterative summary–plan–action–reflection cycles, enabling efficient and interpretable video understanding.
- **Simple yet effective three-stage end to end training pipeline.** Our framework combines SFT cold-start, KTO correction, and GRPO optimization into a scalable pro-

cess that jointly enhances reasoning depth and computational efficiency.

- **High-quality datasets and strong empirical results.** We construct the EVA-SFT, EVA-KTO, and EVA-RL datasets to support stable training, achieving state-of-the-art performance across multiple video benchmarks.

2. Related Works

Agentic Video Understanding. Compared to traditional multimodal large language models (MLLMs) that treat the input video as static context [1, 13, 24, 46], agentic video understanding methods enable MLLM-based agents to actively explore video content using external tools. According to the types of tools employed, existing approaches can be broadly divided into two categories. Ego-R1 [34] and M3-Agent [23] leverage tools that assist in visual comprehension, such as invoking external MLLM APIs or conventional vision models, thereby depending heavily on the tool's performance rather than the base model's inherent multimodal capability. The second category of works [16, 17, 25, 39] equips MLLMs with sampling tools that extract partial or temporal visual information from the video. These methods primarily exploit the agent's planning and recognition abilities, yet still treat the MLLM as a fixed component in a rigid workflow—receiving video input and generating predetermined parameters along a single dimension of control. In contrast, our work restores true autonomy to the agent, enabling it not only to decide which parts of the video to observe, but also how to observe them, with flexible control over spatial resolution and temporal granularity.

Tool-Integrated Reasoning Training. Equipping LLM-based agents with various external tools enables them to interact with the outside world [20, 28, 41], and even to autonomously generate and optimize complex workflows [42, 45]. As foundation models have been trained to produce extended chains of thought for solving complex reasoning tasks [9, 26], recent studies [12, 21] have further integrated tool invocation into the reasoning process and optimized it through reinforcement learning. In this work, we train an MLLM-based agent to iteratively plan and select informative frames, allowing it to flexibly adjust workflows according to the query and the visual content.

3. Method

3.1. Problem Setup

We formulate the active video understanding problem as a Markov Decision Process (MDP). At each timestep t, the agent observes a belief state:

$$s_t = \{q,\, h_t,\, F_t\}, \tag{1}$$

where q denotes the user query, h_t represents the interleaved text–frame history, and F_t corresponds to the visual evidence (frames) obtained from tool calls. The policy of the agent is parameterized as $\pi_\theta(a_t \mid s_t)$.

In video understanding tasks, answering a query does not always require observing uniformly sampled frames. In some cases, such frames are redundant, while in others they fail to provide sufficient evidence for correct reasoning—worse yet, presenting the full video upfront may mislead the planner by anchoring it to spurious or noisy visual cues [16, 27, 36]. Therefore, at the initial step s_0, the model is provided only with the query q, without any visual information in our settings. To enable the agent to autonomously plan its use of visual tokens, we design a flexible *frame-selection tool* that allows both temporal and spatial control.

Parameter	Description
$start\_time$	Start of the temporal window
end_time	End of the temporal window
$nframes$	Number of frames to sample
$resize$	Spatial downsampling ratio

The *start_time* and *end_time* specify the temporal window, while *nframes* denotes the number of frames to sample within this interval. The *resize* parameter enables flexible zoom-in and zoom-out operations.

Intuitively, selecting a larger number of frames enables the agent to better capture dynamic actions, while choosing a higher spatial resolution allows it to extract finer visual details from each frame. This tool schema provides a broad exploration space, encouraging the agent to learn how to allocate temporal and spatial information across rounds to derive precise answers.

Traditional agentic methods can be viewed as constrained instances of our proposed **EVA** framework. They typically employ fixed workflows—such as processing the entire video from the start—and offer limited action freedom (e.g., selecting only a time range). In contrast, EVA can not only execute these human-designed workflows but also dynamically adapt its plan based on the query and the extracted visual evidence, thereby enabling a more general and flexible paradigm for agentic video understanding.

Training such an end-to-end autonomous video agent requires not only visual comprehension and tool-use capabilities but also strong planning skills to determine *what to watch* and *when to answer* based on the question and available visual evidence. Hence, diverse high-quality datasets and efficient training strategies are crucial for developing such a general-purpose agent.

3.2. Data Construction

Supervised Fine-Tuning Data Pipeline We begin by employing Qwen2.5-VL-72B as the teacher MLLM and

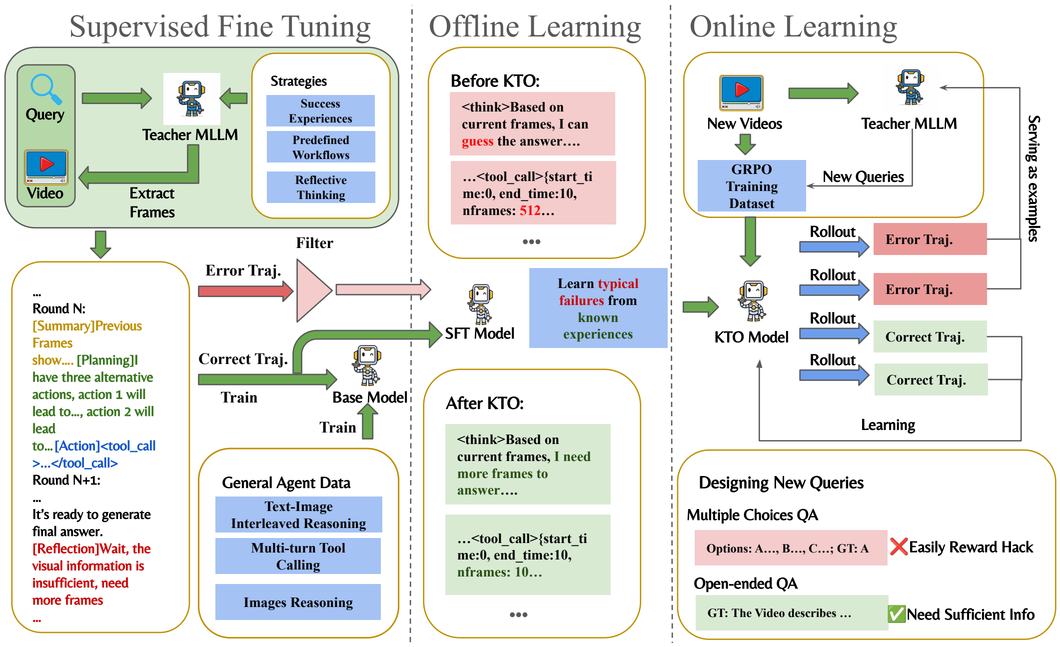

Figure 2. Data Pipeline and Training Stage of EVA. The base model is first fine-tuned on synthetic dataset with certain reasoning and tool-calling pattern. Then we use KTO to help the model learn from typical failures. Finally, we introduce a Data-Enhanced Multi-Stage GRPO training pipeline, where we collect the failure cases of current policy and employ an teacher MLLM to generate new open-ended video QA dataset.

prompting it to generate high-quality agentic video understanding data following our problem setup. The source video QA pairs come from llava-video [46], a short video QA dataset, and cgbench [3], a long video QA dataset. To further enhance data diversity and quality, we design a variety of prompts to guide the teacher model. These include: *Past Success Experiences* generated and summarized by the teacher MLLM itself; *Diverse Workflow Hints* that instruct the model on how to plan and select frames efficiently; and *Reflective Thinking Prompts* that encourage the model to carefully consider its actions.

Inspired by Zhang et al. [43], each SFT data instance follows the format: **Summary + Planning + Action + Reflection**. In the **Summary** stage, the MLLM generates a detailed description of the content for each frame, which explicitly pushes the model to attend to the returned visual evidence and thereby better ground its understanding of the tool's parameters and outputs. During **Planning**, since an autonomous video agent possesses maximum flexibility to select actions from an extremely large action space, it is crucial to train its ability to propose potential actions based on current information while estimating their cost and outcome. In the **Action** stage, the model generates appropriate tool calls. Finally, as models tend to produce answers without sufficient visual evidence, resulting in degraded performance, we construct **Reflection** data that guide the model to evaluate whether the available visual information is adequate before producing an answer. If not, the model continues to call tools to gather additional information.

Kahneman-Tversky Optimization Data Pipeline The SFT-trained model effectively learns tool-calling formats and reasoning patterns; however, it still struggles to select appropriate strategies. Several typical failure cases share similar patterns: the model may generate answers without sufficient visual evidence, sample too many frames within a short temporal window, or too few frames across a relatively long one. To address these recurrent failures and stabilize subsequent online reinforcement learning, we employ the KTO framework. Unlike DPO [30], which requires pairwise preference data and thus enforces a shared dialogue round—an assumption misaligned with our multi-turn interaction setting that may truncate strategies—KTO only requires single-sample preference labels ("chosen" or "rejected"). Compared with GRPO, KTO further enables the model to learn from externally collected experiences rather than self-play, leading to a more stable and sample-efficient training process. Specifically, we collect incorrect trajectories from the SFT data construction pipeline and categorize them as rejected samples. The criterion of data selection are two folds. Firstly, we use LLM As Judge to select the trajectories whose reasoning process shows it do not have enough visual tokens but still generate a answer, representing for guessing. Secondly, We also resample high-quality

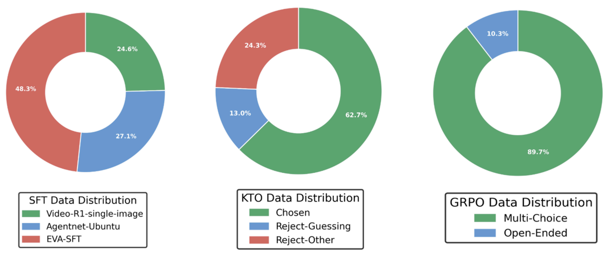

Figure 3. Distribution of the Training Dataset

successful trajectories as chosen data. This setup allows the model to learn fine-grained preferences between fully successful and failed trajectories. Figure 2 illustrates representative examples before and after applying KTO.

GRPO Data Pipeline. GRPO is an online reinforcement learning framework in which the model generates multiple rollouts by itself and iteratively learns from both successes and failures. However, conventional GRPO typically relies on a static training dataset, and the model only iterates through it for a few epochs. This limitation becomes more pronounced when training a video agent. Traditional GRPO learns from failures based solely on a fixed query–video pair, which constrains the diversity of challenges encountered. For instance, the model may realize that its counting ability is weak, yet it can only improve from a limited set of failed queries and videos.

To address this issue, we introduce a Data-Enhanced GRPO pipeline. We first construct a reinforcement learning dataset by collecting failure cases from the KTO-trained model. After several GRPO training steps, we further gather new failure cases and use them as in-context examples for the teacher MLLM, which then generates new question–answer pairs for unseen videos from HD-VILA [40], conditioned on those examples. And We will re-train GRPO model on the enhanced dataset.

Specifically, the teacher MLLM is prompted to produce open-ended QA pairs with concise answers. Compared to directly designing multiple-choice questions, this open-ended formulation mitigates reward hacking caused by answer guessing and offers a more efficient generation process for the teacher model, since designing balanced multiple-choice options often introduces unintended information cues and additional complexity.

3.3. Reinforcement Learning

We optimize the model via reinforcement learning using a mixed-format dataset. Specifically, we train EVA on both multiple-choice and open-ended questions. For multiple-choice questions, we adopt the Completeness Self-Verification (CSV) reward [27] to ensure that EVA explicitly identifies the correct frames rather than relying on random guessing. For open-ended questions, we utilize the ROUGE score as the reward signal.

3.3.1. GRPO Primer

We employ *Group Relative Policy Optimization* (GRPO) [31], a KL-regularized policy optimization method that encourages high-return behaviors while constraining the policy to remain close to a reference model π_{ref} initialized via SFT and KTO. The training objective is formulated as:

$$\max_{\theta} \; \mathbb{E}_{\tau\sim\pi_\theta}\big[R(\tau)\big] - \lambda\,\mathbb{E}_{(s,a)\sim\pi_\theta}\big[\mathrm{KL}(\pi_\theta(\cdot\mid s)\,\|\,\pi_{\text{ref}}(\cdot\mid s))\big]\,. \tag{2}$$

3.3.2. Reward Shaping

Accuracy Reward. Our GRPO training corpus contains both *open-ended* and *multiple-choice* question–answer pairs. Accordingly, we design a composite reward function that adapts to these two types of supervision:

$$R(\tau) = w_{\text{acc}}\, r_{\text{acc}} + w_{\text{fmt}}\, r_{\text{fmt}}. \tag{3}$$

The accuracy reward is defined as:

$$r_{\text{acc}} = \begin{cases} r_{\text{csv}}, & \text{if multiple-choice},\\ r_{\text{rouge}}, & \text{if open-ended}. \end{cases} \tag{4}$$

Specifically, for multiple-choice tasks, we set up the same base model to act as a judge, feeding it the question alongside EVA's last round of retrieved images. We set $r_{\text{csv}} = 1$ if and only if both the judge and EVA produce the correct answer; otherwise, $r_{\text{csv}} = 0$.

For open-ended tasks, let $R_1, R_2, R_L \in [0,1]$ denote the ROUGE-1, ROUGE-2, and ROUGE-L F1 scores between the generated answer and the ground truth (with stemming). The averaged ROUGE reward is then defined as:

$$r_{\text{rouge}} = \tfrac{1}{3}\big(R_1 + R_2 + R_L\big) \in [0,1]. \tag{5}$$

Format Reward. We also introduce a format reward to prevent the model from directly guessing the answer without proper reasoning. Specifically, if the model generates a tool call but ultimately yields an incorrect answer, we provide a compensatory format reward of 0.05. Considering that the expected accuracy for random guessing is approximately 0.20 or 0.25 (depending on the number of choices), this deliberately low reward discourages the model from exploiting the formatting structure to gain undeserved points through random guessing.

4. Experiments

4.1. Experiment Settings

We choose Qwen2.5-VL-7B-Instruct as our base model, as it supports vision input at various resolution and save token when feeding frames with small resolutions.

We first perform SFT using our EVA-SFT data and open-sourced agentic training data. The model is trained for 2 epoch, with batch size = 8 and learning rate = $2e-6$. The model is then trained using KTO [10] using EVA-KTO data. Follow the recommended chosen and reject data proportion, there are 63% correct trajectory and 37% incorrect data in the training dataset. We keep the same learning rate and set $beta = 0.1$. We further train our model using GRPO based on our EVA-RL data, with 90% open-ended QA and 10% multiple-choice questions (MCQ). We use a combined reward as described in section 3.3.2. In detail, multiple choice questions are rewarded for correct choice and open-ended questions are rewarded using rouge-score. The model is trained for 1 epoch, with batch size =64, number of rollout per sample =8 and learning rate $1e-6$, on 32 H100 GPUs.

We evaluate EVA on various video benchmarks including LSDBench [29], LongVideoBench [38], MLVU [47], VideoMME [14], LVBench [37] and Video-Holmes [8]. The metrics for all benchmarks are accuracy, computed as the proportion of correctly predicted answers.

4.2. Main Result

Sampling Dilemma Bench. We first evaluate our model on the Sampling Dilemma Bench (LSDBench [29]) to examine its ability to balance sampling efficiency and visual understanding accuracy. As shown in Table 1, closed-source models such as Gemini-2.0-Flash achieve the highest accuracy (56.2%) but rely on an extremely large number of visual tokens (over 700K), revealing the inefficiency of brute-force dense sampling. Among open-source models, Qwen2.5-VL and InternVideo2.5 achieve comparable results around 50% using 256–768 frames. Building upon Qwen2.5-VL, we introduce an end-to-end video agent that performs planning-before-perception via iterative reasoning, tool calls, and reflection. This design allows the model to dynamically decide which frames to observe and reason over, rather than exhaustively processing all inputs. As shown in Table 1, EVA exhibits clear improvements, achieving 51.8% with only 6.2K visual tokens, surpassing the baseline by +2.6% while using significantly fewer tokens. These results demonstrate that our video agent effectively mitigates the sampling dilemma through reasoning-driven visual planning, enabling more efficient video understanding.

Long-Form Video Understanding. We further evaluate EVA on four long-form video benchmarks—LongVideoBench, MLVU, VideoMME, and LVBench—to assess its generalization in temporally extended scenarios. As shown in Table 2, EVA achieves strong and consistent results (55.1%, 60.5%, 59.9% and 38.1%), outperforming most open-source and adaptive

Table 1. Performance on sampling dilemma bench. We report different stages of EVA model's performance on LSDbench. SOTA model performance are directly from [17]. The visual token are roughly measured by 258, 144 ,256 and 650 token per frame for gemini, LongVA, LongVila and Qwen-VL family models.

Method	Frames	Visual Token	Acc (%)
Close Source Model			
Gemini-2.0-Flash [33]	2700	696.6k	56.2
Open Source Model			
LongVA [44]	256	36.9k	31.3
	512	73.7k	33.0
	1024	147.5k	32.5
Qwen2-VL [35]	256	166.4k	48.0
Qwen2.5-VL [2]	256	166.4k	50.1
	768	499.2k	52.5
LongVila [5]	256	65.5k	49.8
Qwen2.5-VL(RHS) [17]	225	146.2k	52.2
Baseline			
Qwen2.5-VL*	16	10.5k	47.7
	32	21.0k	49.2
Ours			
EVA	76.9	10.3k	**51.0**

agents while processing only about 20–30 frames per video. The number of frames is estimated by assuming 650 visual tokens per frame for fair comparison; in practice, EVA may use more frames with adaptively adjusted resolutions. This demonstrates the advantage of our planning-before-perception paradigm in long-form video understanding, where the video agent adaptively decides which segments to observe through iterative reasoning and reflection. By dynamically allocating attention rather than relying on fixed dense sampling, EVA maintains high accuracy with minimal visual tokens, effectively addressing the long-context challenge in video understanding.

Zero-Shot Performance on Video Reasoning Bench. We further evaluate EVA on the Video-Holmes benchmark, which assesses diverse reasoning abilities. As shown in Table 3, despite being evaluated in a zero-shot setting, EVA achieves competitive overall performance (32.6%, 32.9% and 36.7% for EVAs), comparable to open-source models such as Video-R1 and VideoChat-R1 with uniformly sampled frame schema. This demonstrates the strong transferability of our reasoning-driven video agent: even without task-specific supervision, EVA can generalize to multi-step reasoning and causal understanding across long temporal contexts. The results highlight that our planning-before-perception paradigm not only improves efficiency but also enables robust general reasoning in video understanding.

4.3. Ablation Study

Training Schema Our ablation study confirms the SFT–KTO–GRPO sequence provides a clear evolutionary path for EVA. As shown in Fig. 4 and Table 2, the SFT

Table 2. Main performance on multiple video understanding benchmark. Baseline results are directly cited from [17].The number of frames for EVA, which is indicated by *, is estimated by assuming 650 visual tokens per frame for fair comparison; the actual number of frames may vary depending on the resolution determined by the model adaptively.

Model	LongVideoBench		MLVU		VideoMME-Long/Overall		LVBench	
	Frame	Acc	Frame	Acc	Frame	Acc	Frame	Acc
Close Source Models								
GPT-4o [19]	32	58.2	0.5 fps	64.6	384	65.3/71.9	60	48.9
Gemini-1.5-Pro [33]	32	55.2	-	-	0.5fps	67.4/75.0	3600	33.1
Static Frame Sampling								
ShareGPT4Video [4]	16	39.7	16	46.4	16	35.0/39.9	-	-
LongVA [44]	-	-	256	56.3	128	46.2/52.6	-	-
VITA-1.5-7B [15]	-	-	-	-	16	47.1/56.1	-	-
Video-R1 [13]	32	52.7	32	60.2	32	49.4/59.9	32	35.3
VideoChat-R1 [22]	32	49.1	32	54.3	32	46.2/-	32	34.3
Qwen2.5-VL [2]	32	43.2	32	48.4	32	44.7/53.6	32	31.6
Adaptive Agent								
VideoAgent [11]	-	-	-	-	87	49.0/56.0	25.5	29.3
FrameThinker [17]	21.1	52.9	23.2	59.1	24.1	47.6/-	23.9	36.6
VideoMTR [39]	-	-	-	-	32	51.0/59.0	-	-
Ours								
EVA-SFT	33.8*	49.9	46.7*	52.3	26.6*	45.8/56.0	56.2*	26.5
EVA-KTO	35.6*	53.2	28.7*	57.4	24.1*	45.1/56.5	34.5*	36.0
EVA-GRPO	25.3*	**55.0**	22.2*	**68.3**	22.8*	**48.4/60.2**	26.8*	**43.3**

Table 3. Zero-shot performance on video reasoning benchmark: Video-Holmes [8], where SR stands for Social Reasoning; IMC stands for Intention & Motive Chaining; TCI stands for Temporal Causal Inference; TA Timeline Analysis; MHR stands for Multimodal Hint Reasoning; PAR stands for Physical Anomaly Reasoning; CTI stands for Core Theme Inference.

Model	Frame	SR	IMC	TCI	TA	MHR	PAR	CTI	Overall
Close Source Models									
GPT-4o [19]	32	50.0	49.6	38.8	30.0	44.0	39.2	37.0	42.0
Gemini-2.0-Flash [33]	-	41.8	33.7	23.1	20.5	30.1	26.8	33.7	30.6
Open Source Model									
InternVL2.5-8B [7]	32	28.0	32.2	21.5	7.7	25.7	23.8	22.6	23.8
InternVL3-8B [48]	32	29.5	40.7	37.9	35.1	24.6	38.9	24.1	32.3
Qwen2.5-VL-7B [2]	32	38.4	34.8	17.6	30.0	27.1	18.6	25.2	27.8
SEED-Bench-R1 [6]	32	42.8	35.1	25.6	40.5	29.2	29.9	32.6	33.5
VideoChat-R1 [22]	32	42.1	38.8	24.5	39.5	29.5	27.8	29.3	33.0
Video-R1 [13]	32	48.6	**41.7**	**28.9**	34.5	**31.0**	33.6	35.6	36.5
Ours									
EVA-SFT	11.5*	44.5	33.7	26.4	39.5	23.2	31.9	32.2	32.6
EVA-KTO	5.8*	48.6	36.2	22.7	39.5	22.9	32.0	31.1	32.9
EVA-GRPO	36.8*	**49.3**	39.5	**30.4**	**44.5**	27.1	37.6	**35.2**	**37.2**

model consumes a large number of frames and rounds yet achieves the lowest performance, indicating that supervised fine-tuning alone teaches the agent to follow tool-calling formats but not to explore videos efficiently. The KTO model significantly reduces both frame consumption and interaction rounds while delivering a substantial improvement over SFT. Interestingly, the GRPO model further reduces the number of sampled frames compared to KTO, yet increases the number of interaction rounds, and achieves the highest scores across all benchmarks. This reveals a shift in exploration strategy: rather than passively consuming fewer frames in fewer steps, the GRPO-trained agent learns to engage in more deliberate, multi-round reasoning while allocating its visual token budget more precisely in each round. These results illustrate that our training scheme progressively transforms the Video Agent from a format-following

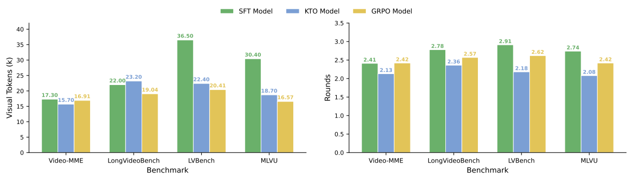

Figure 4. Distribution of Rounds and Visual Token cross Models and Benchmarks

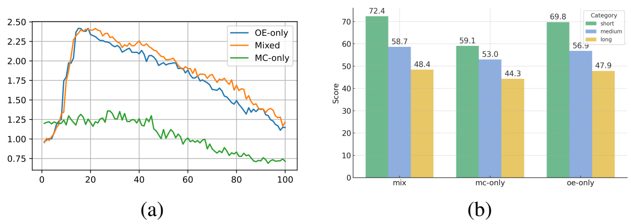

Figure 5. Ablation study on the GRPO training dataset. The comparison between multi-choice (MC) only, open-ended (OE) only, and mixed (MC+OE) data shows that mixed data provides a more effective learning environment for the agent, which leads to better performance on VideoMME.

imitator into a strategic video explorer that actively plans when and where to look.

Data Composition in GRPO Although KTO effectively encourages broader exploration, we observe that the model may still hack the reward by producing plausible but visually unsupported guesses when limited visual evidence is available. To mitigate this issue, we introduce open-ended data during GRPO, forcing the agent to ground its answers in the visual content even when exploring diverse trajectories. As illustrated in Fig. 5, mixing open-ended data with task-specific data leads to more stable training and clearly improved performance, demonstrating that proper data composition is crucial for preventing reward hacking and ensuring visually consistent reasoning.

4.4. Computation Efficiency Compared with Uniform Sampling Schema

Although our agent performs multi-round planning and perception, its overall computation remains highly efficient. As shown in Table 1, the total number of tokens is comparable to or even lower than static uniform-sampling baselines, with text tokens occupying only a small fraction of the computation budget. Consequently, the inference runtime does not scale with the number of reasoning steps but is instead dominated by the compact set of adaptively selected visual tokens. This allows our agent to achieve competitive or superior efficiency compared with uniform sampling, despite operating under a more expressive reasoning framework.

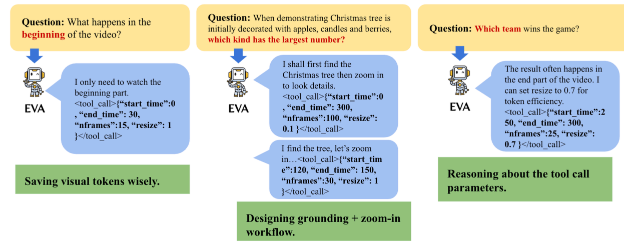

Figure 6. Diverse Workflows generated by EVA.

4.5. Case Study

Figure 6 illustrates the diverse workflows produced by EVA for different types of questions. Many queries require visual information from only a specific segment of the video (e.g., the beginning or the end). Under the plan-before-perception scheme, EVA allocates visual tokens precisely where needed, avoiding redundant usage that often occurs in traditional MLLM or agent methods, which perceive before reasoning or tool-calling.

When a query genuinely requires extensive visual evidence, EVA can also behave like conventional video agents—first sampling uniformly for grounding and then zooming in on key frames to extract finer details, showing EVA is a more general video agent.

The larger action space further differentiates EVA from existing Video Agents. For example, when performing a zoom-in operation, prior agents can only adjust the temporal range but cannot vary both the number of sampled frames and the spatial resolution. In contrast, EVA autonomously selects all relevant parameters, enabling workflows that not only cover those of other agents but also surpass them through more efficient visual token usage.

5. Conclusion and Limitation

In this work, we take a step toward building truly autonomous video-understanding agents by introducing a query-driven framework that integrates understanding, planning, tool use, and reflection in an iterative loop. Through a three-stage training paradigm—SFT cold-start, KTO, and GRPO with a commonly used reward—the model learns to balance perceptual efficiency and reasoning depth, progressively evolving from a passive video recognizer into an adaptive and self-directed agentic watcher. Despite these promising results, our approach still faces several limitations. The current reasoning loop relies on pre-defined tool interfaces and may struggle with unseen or noisy query distributions. Future work will explore more flexible tool ecosystems, self-evolving reasoning strategies, and cross-modal memory mechanisms to further enhance autonomy and generalization in long-horizon video understanding.

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